

D302

Implementing and Evaluating E-Learning Solutions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A new e-learning module for adult learners at a regional credit union included six practice scenarios with feedback. Logs show most sites skipped the scenarios to save time, and post-test scores did not improve. The credit union recently opened two branches. What should the evaluator conclude?

- A. The module does not work and should be dropped.
- B. The module was not implemented with fidelity, so its effectiveness cannot yet be judged.
- C. The new branches caused the flat scores.
- D. Add more scenarios to the module.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Scores rose sharply after a module launched. What should an evaluator check before crediting the design?

- A. Whether scores fell at other sites.
- B. Whether the module was delivered as designed, so the gain can be tied to the design itself.
- C. Nothing; good results mean the design worked.
- D. Whether learners liked the module.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A designer has finished an untested e-learning module for 4,000 adult employees across 12 offices. The chief learning officer is eager to launch before a holiday. The module has an attractive, modern look. What is the best implementation plan?

- A. Launch to all 4,000 before the holiday.
- B. Launch to all 12 offices, then fix problems as complaints arrive.
- C. Never launch until the module is perfect.
- D. Pilot it with one office, fix problems, then scale.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Adult learners who used a new module averaged 80 on a skills test; a comparison group averaged 72. The pooled standard deviation was 16. The study ran in a large office park with free parking. What is Cohen's d?

- A. 0.11
- B. 8
- C. 0.50
- D. 2.00

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A study reports $d = 0.25$ with a pooled standard deviation of 12. The comparison group mean was 70. What was the treatment group mean?

- A. 70.25
- B. 67
- C. 73
- D. 82

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A designer concludes that an e-learning module improved adult learners' confidence and job performance, based only on an end-of-course self-report survey with a 92% response rate. The survey was built in a popular free tool. How should the conclusion be strengthened?

- A. Add other sources, such as supervisor observations and work performance data, to check the self-reports.
- B. Rebuild the survey in a paid tool.
- C. Add more questions to the same survey.
- D. Raise the response rate above 92%.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Dashboard data show adult learners who opened an optional practice tool more than ten times scored 15 points higher than those who used it less. Learners chose how often to use it. The tool was developed by a partner university. What can the designer conclude?

- A. The tool works because a university developed it.
- B. The tool caused a 15-point gain.
- C. The tool has no effect at all.
- D. Tool use and scores are associated, but motivated learners may both use it more and score higher.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A designer wants to know whether an optional practice tool causes higher scores. Which design best supports a causal claim?

- A. Survey users about whether they think the tool helped.
- B. Compare heavy and light users as they chose.
- C. Track tool use for a longer period.
- D. Randomly assign learners to use the tool or not, then compare scores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A post-test for an adult e-learning module has three sections. Learners averaged 88% on Section 1 (Objective 1), 85% on Section 2 (Objective 2), and 41% on Section 3 (Objective 3). The sponsor wants a new opening video. The module was translated into two languages. What revision does the data support?

- A. Redesign the whole module.
- B. Revise the instruction and practice for Objective 3.
- C. Revise Objective 2, since 85% is below Objective 1.
- D. Add a new opening video.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

After revising only Objective 3's lesson, Section 3 scores rose from 41% to 79% while the other sections stayed about the same. What does this suggest?

- A.** The targeted revision likely addressed the Objective 3 weakness.
- B.** The gain is 79 points.
- C.** The other sections should be revised too, since they did not improve.
- D.** The whole module is now fixed and needs no more testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

At the end of a design-based study, a designer writes the reflect section. The project ran for two terms at a nonprofit adult literacy center with a new building. Which reflection best fits design-based research?

- A.** Short, spaced practice with instant feedback helped adult readers with limited time; designers in similar settings should build practice in short blocks.
- B.** I felt proud of the project and enjoyed working with the staff.
- C.** This design will work for every adult learner everywhere.
- D.** The new building helped the project succeed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A designer's study found the module improved quiz scores but had no effect on on-the-job error rates. The final report has a polished layout and a two-page executive summary. The designer's manager suggests leaving out the error-rate finding. What should the designer do?

- A.** Shorten the executive summary to one page.
- B.** Leave out the error-rate finding as suggested.
- C.** Report both findings and discuss limitations and possible reasons for the null result.
- D.** Report the error rates as improved, since quiz scores improved.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A new e-learning module for adult learners at a regional credit union included six practice scenarios with feedback. Logs show most sites skipped the scenarios to save time, and post-test scores did not improve. ~~The credit union recently opened two branches.~~ What should the evaluator conclude?

Target word: conclusion given skipped components **Decides:** The practice scenarios, a core part of the design, were skipped.

A	Absolute language	The result reflects poor implementation, not a fair test.
B	Correct	You cannot evaluate a design that was not delivered as designed.
C	Camouflage	An irrelevant organizational detail.
D	Inverted operation	Adds more of what was already being skipped.

2. Answer B Concept 1 · transfer

Scores rose sharply after a module launched. What should an evaluator check before crediting the design?

Target word: reverse: fidelity with good results **Decides:** Whether the design was actually delivered as planned.

A	Confusion set	A comparison can help, but it does not check fidelity.
B	Correct	If delivery changed, the gain may reflect something else.
C	Absolute language	Results must be linked to what was actually delivered.
D	Adjacent category	Satisfaction does not show what was delivered.

3. Answer D Concept 2

A designer has finished an untested e-learning module for 4,000 adult employees across 12 offices. The chief learning officer is eager to launch before a holiday. ~~The module has an attractive, modern look.~~ What is the best implementation plan?

Target word: rollout approach **Decides:** The module is untested.

A	Camouflage	The deadline pressure does not change the need to test.
B	Inverted operation	Tests on everyone instead of a small group.
C	Absolute language	Perfection is not the standard; a pilot is.
D	Correct	A small pilot catches issues before they affect everyone.

4. Answer C Concept 3

Adult learners who used a new module averaged 80 on a skills test; a comparison group averaged 72. The pooled standard deviation was 16. ~~The study ran in a large office park with free parking.~~ What is Cohen's d?

Target word: effect size **Decides:** $d = (80 - 72) / 16$.

A	Wrong stem number	Divides the difference by the comparison mean (8/72) instead of the SD.
B	Dropped element	The raw difference without dividing by the SD.
C	Correct	$8 / 16 = 0.50$, a moderate effect.
D	Inverted operation	Divides the SD by the difference (16/8).

5. Answer C Concept 3 · transfer

A study reports $d = 0.25$ with a pooled standard deviation of 12. The comparison group mean was 70. What was the treatment group mean?

Target word: solve for a mean **Decides:** Difference = $d \times SD = 3$.

A	Dropped element	Adds d without multiplying by the SD.
B	Inverted operation	Subtracts the difference.
C	Correct	$0.25 \times 12 = 3$; $70 + 3 = 73$.
D	Wrong stem number	Adds the SD instead of $d \times SD$.

6. Answer A Concept 4

A designer concludes that an e-learning module improved adult learners' confidence and job performance, based only on an end-of-course self-report survey with a 92% response rate. ~~The survey was built in a popular free tool.~~ How should the conclusion be strengthened?

Target word: strengthen the evidence **Decides:** A single self-report source.

A	Correct	Triangulation confirms findings across sources.
B	Camouflage	The tool is not the weakness.
C	Adjacent category	Still one self-report source.
D	Wrong stem number	More of the same source does not triangulate.

7. Answer D Concept 5

Dashboard data show adult learners who opened an optional practice tool more than ten times scored 15 points higher than those who used it less. Learners chose how often to use it. ~~The tool was developed by a partner university.~~ What can the designer conclude?

Target word: valid inference from association **Decides:** Learners self-selected into use.

A	Camouflage	Source prestige does not show cause.
B	Confusion set	Treats association as causation.
C	Absolute language	The data cannot rule out an effect either.
D	Correct	Self-selection means another factor could explain both.

8. Answer D Concept 5 · transfer

A designer wants to know whether an optional practice tool causes higher scores. Which design best supports a causal claim?

Target word: reverse: design for causation **Decides:** Random assignment removes self-selection.

A	Adjacent category	Perceptions are not causal evidence.
B	Inverted operation	Keeps the self-selection problem.
C	Dropped element	More time does not remove the comparison problem.
D	Correct	Randomization balances motivation and other factors.

9. Answer B Concept 6

A post-test for an adult e-learning module has three sections. Learners averaged 88% on Section 1 (Objective 1), 85% on Section 2 (Objective 2), and 41% on Section 3 (Objective 3). ~~The sponsor wants a new opening video. The module was translated into two languages.~~ What revision does the data support?

Target word: data-driven revision **Decides:** Section 3 is far lower than the others.

A	Absolute language	Two objectives perform well; a full redesign is not supported.
B	Correct	The data point clearly to Objective 3.
C	Wrong stem number	A small difference; Objective 3 is the outlier.
D	Camouflage	A sponsor preference, not what the data show.

10. Answer A Concept 6 · transfer

After revising only Objective 3's lesson, Section 3 scores rose from 41% to 79% while the other sections stayed about the same. What does this suggest?

Target word: reverse: interpret the result of a targeted revision **Decides:** Only the revised section changed.

A	Correct	The change appeared only where the revision was made.
B	Wrong stem number	The gain is 38 points (79 – 41).
C	Inverted operation	They were already strong and were not revised.
D	Absolute language	Iteration continues; one result does not end the cycle.

11. Answer A Concept 7

At the end of a design-based study, a designer writes the reflect section. The project ran for two terms at a nonprofit adult literacy center ~~with a new building~~. Which reflection best fits design-based research?

Target word: purpose of reflection in DBR **Decides:** Reflection should produce transferable design principles.

A	Correct	A reusable principle tied to context and evidence.
B	Adjacent category	Personal feelings, not a design principle.
C	Absolute language	Principles are bounded by context.
D	Camouflage	Not a design lesson from the data.

12. Answer C Concept 8

A designer's study found the module improved quiz scores but had no effect on on-the-job error rates. ~~The final report has a polished layout and a two-page executive summary.~~ The designer's manager suggests leaving out the error-rate finding. What should the designer do?

Target word: ethical reporting **Decides:** An unfavorable finding is being dropped.

A	Camouflage	Format is not the issue.
B	Dropped element	Omits a real result.
C	Correct	Honest reporting includes null results and limitations.
D	Inverted operation	Misstates the finding.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	2	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Implementation fidelity first	Before judging whether a solution works, check whether it was delivered as designed.	Concluding a design failed when it was never implemented as planned.	Weak results plus signs that parts were skipped or changed.
2	Pilot before full rollout	Test the implementation with a small group to catch problems before scaling.	Launching to everyone at once to get more data faster.	A first release decision for an untested solution.
3	Effect size, not just significance	Effect size (e.g., Cohen's d = mean difference / standard deviation) shows how large a difference is; a p-value only shows whether it is likely due to chance.	Treating a significant result as a large or important one.	Two means and a standard deviation, or a tiny p-value on a huge sample.
4	Triangulation	Confirm findings with several data sources or methods so one flawed source does not drive conclusions.	Relying on a single measure, often self-report.	A conclusion resting on one survey.
5	Correlation is not causation	An association between use and outcomes does not show the solution caused them; other factors may drive both.	Claiming heavy users did better because of the tool.	Users self-selected into more or less use.
6	Let the data point to the revision	Revise the part of the design linked to the weakest results (e.g., the objective behind the most-missed items).	Redesigning everything or the most visible feature.	Item or section results that differ sharply.
7	Reflection yields design principles	Design-based research ends with reflection that turns what happened into principles others can reuse in similar settings.	Thinking reflection is just a personal feelings journal.	The reflect phase of the study.
8	Honest reporting	Report all findings, including null or negative results, along with limitations.	Leaving out results that did not support the design.	A report draft that drops an unfavorable finding.

Course scope (public description): Implementing and Evaluating E-Learning Solutions is the third of three capstone courses in the program. This course focuses on the final steps of the Design-Based Research process: implement, test, refine, reflect, and report. This course also requires that learners continue applying Design Thinking as they test and refine the solution identified during the prototyping phase. The course requires learners to test and refine their implementation strategies, use data to evaluate the effectiveness of their e-learning solution, redesign or enhance their e-learning design based on their interpretation of the data, and summarize their design-based action research study. Developing an E- Learning Solution and Research Methodology is a prerequisite for this course.